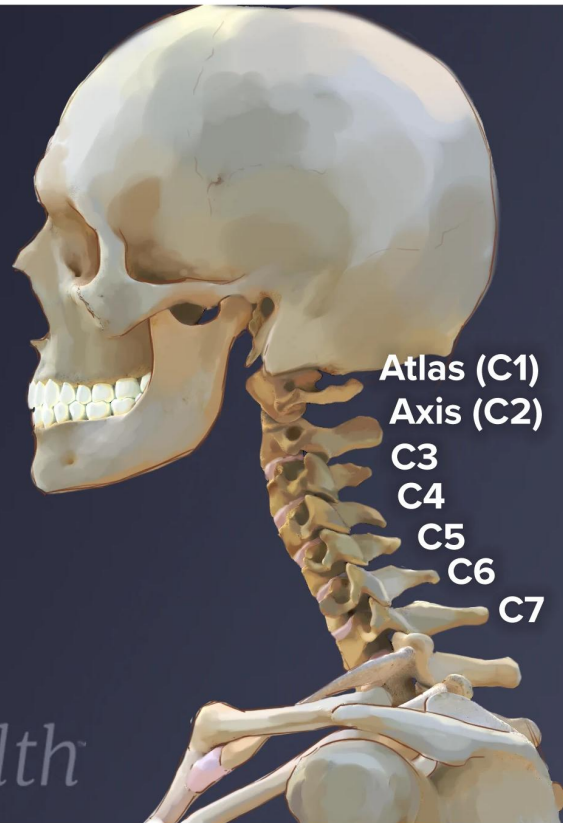
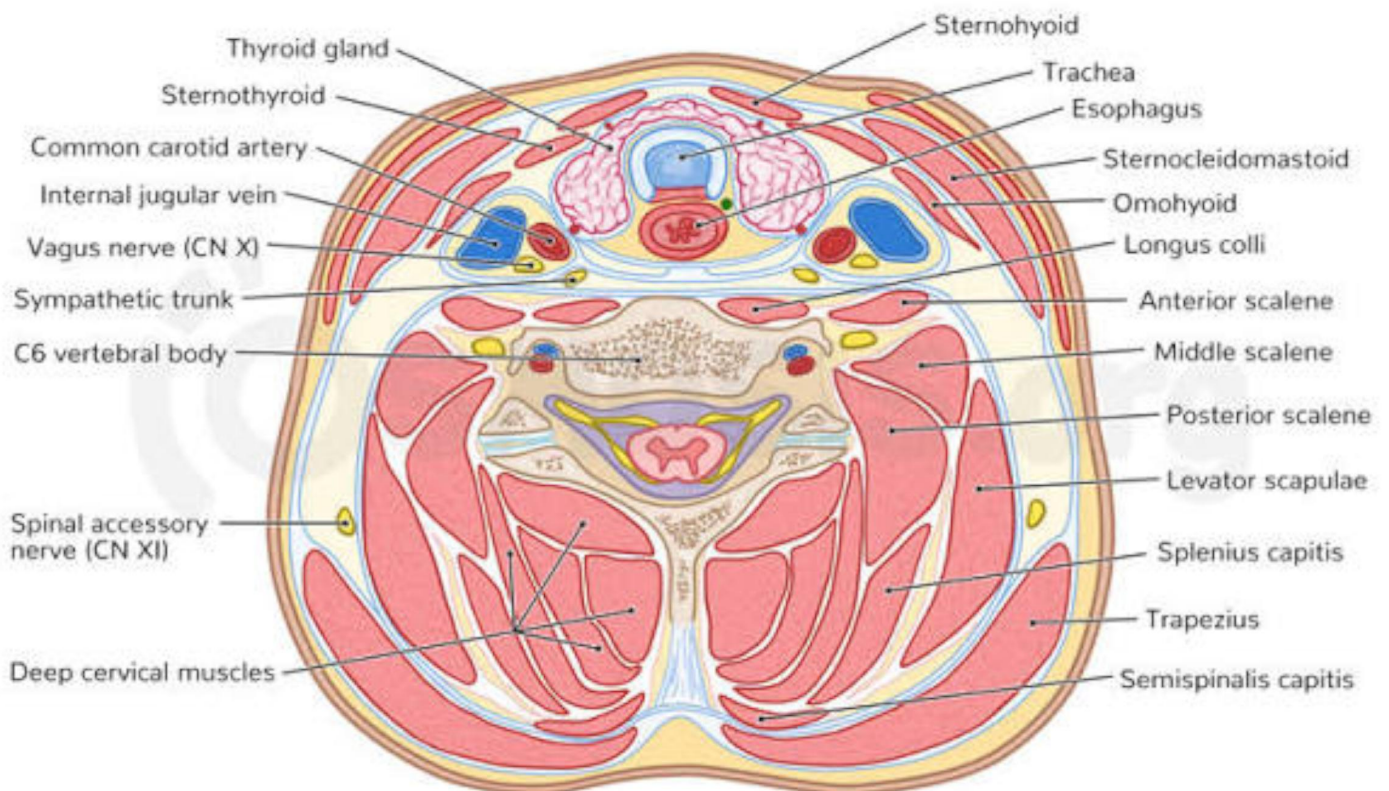


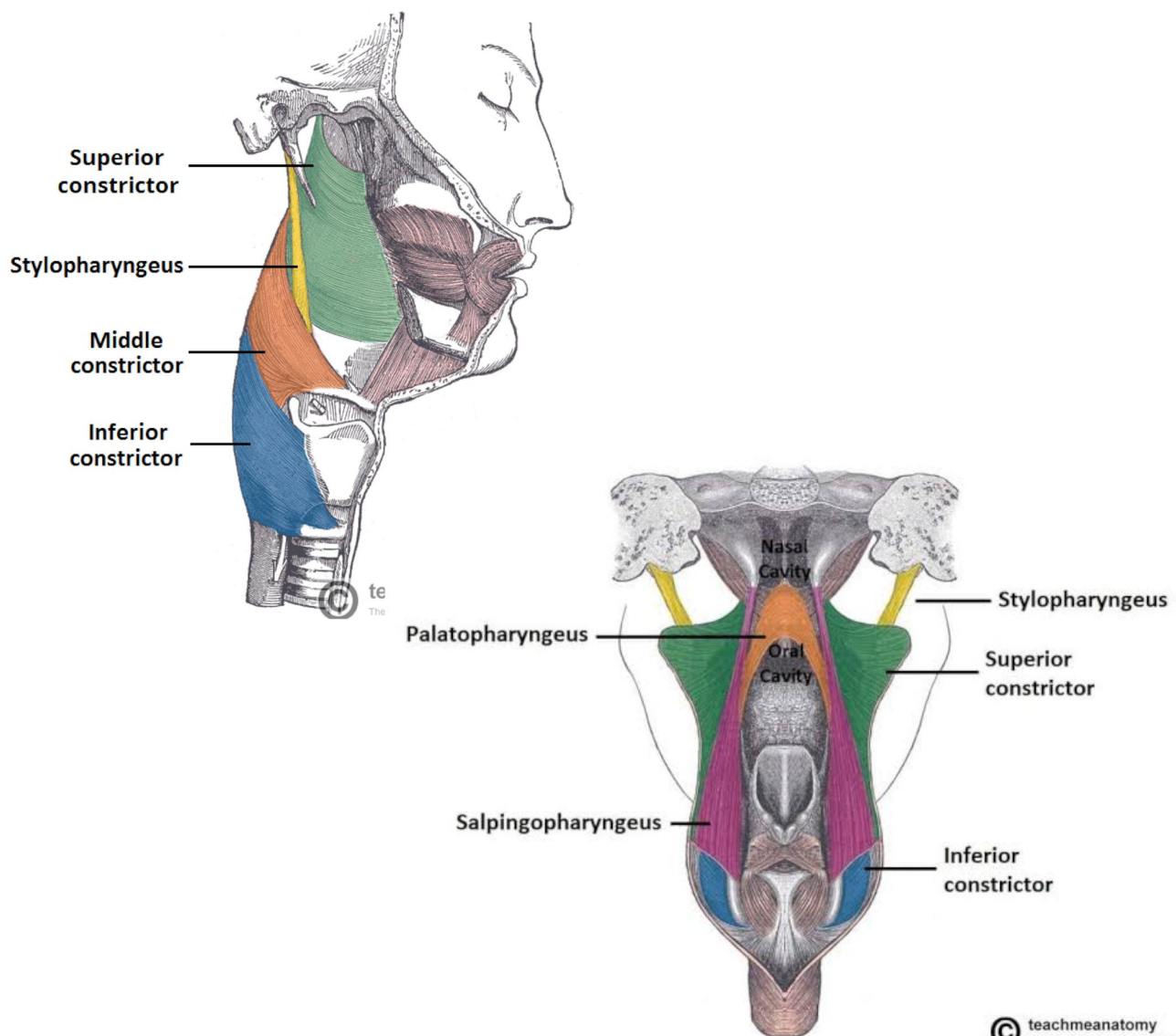
THE NECK

CERVICAL SPINE



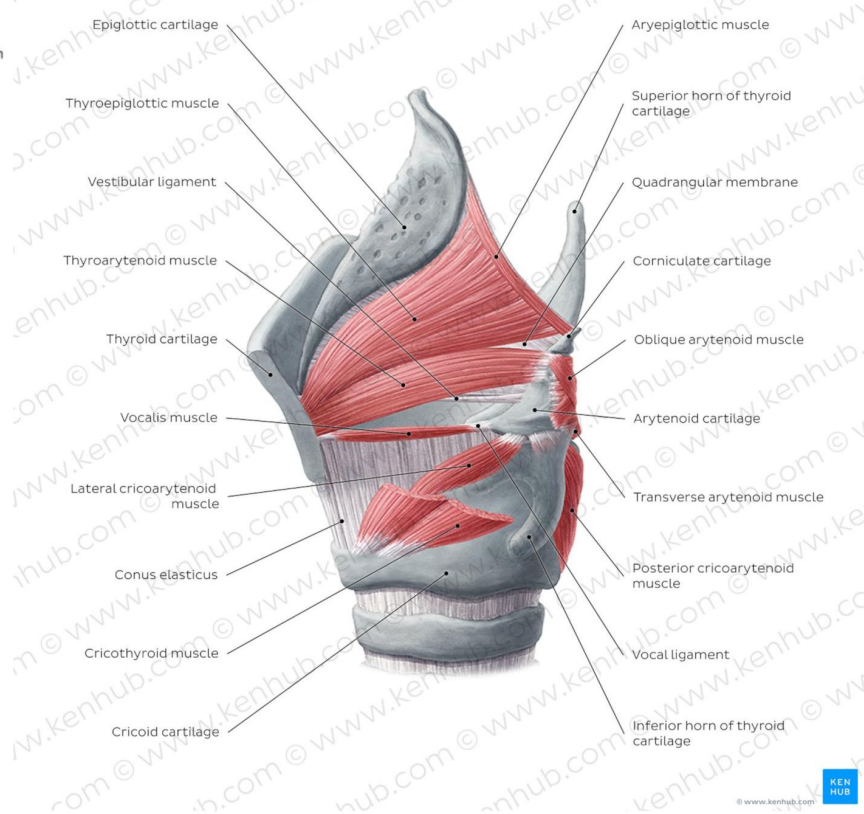
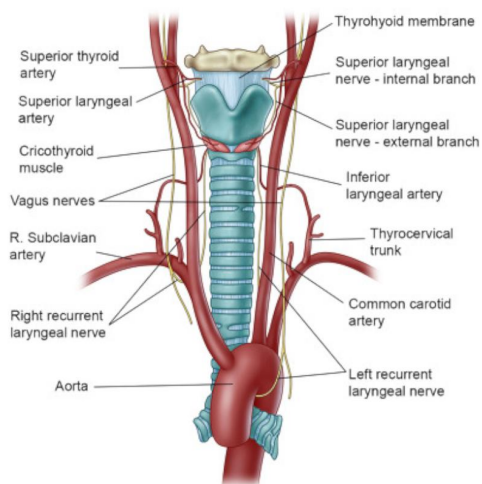
VERITAS *health*



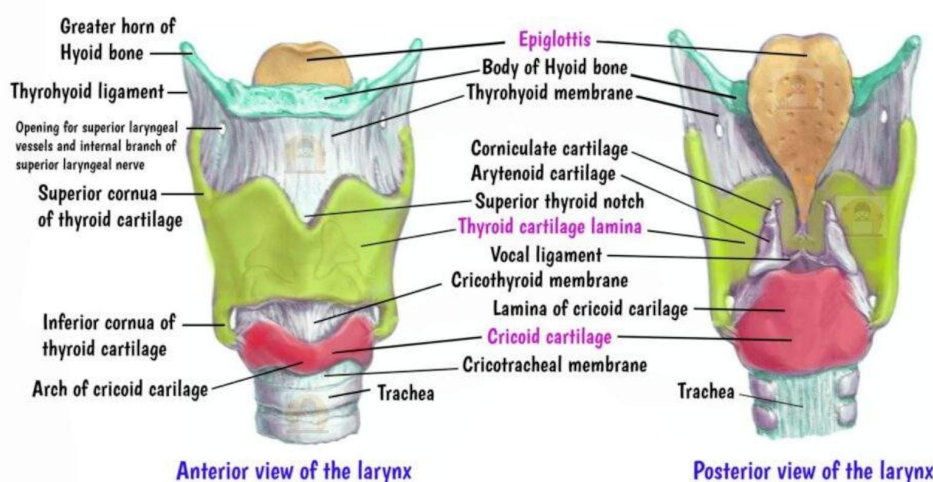


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The pharyngeal muscles form the muscular wall of the pharynx, a cone-shaped tube located in the neck, positioned posterior to the nasal and oral cavities and extending down to the larynx and esophagus. This muscular layer is composed of three paired, overlapping constrictor muscles—the superior, middle, and inferior pharyngeal constrictors—which are arranged like stacked cups. These constrictors anchor anteriorly to various bony and cartilaginous structures of the skull base, mandible, hyoid bone, and thyroid and cricoid cartilages, wrapping posteriorly to meet at a fibrous midline seam called the pharyngeal raphe. Their primary action is sequential, wavelike contraction that propels a bolus of food or liquid downward from the pharynx into the esophagus during swallowing, a process known as peristalsis. Complementing them are the longitudinal pharyngeal muscles—the stylopharyngeus, palatopharyngeus, and salpingopharyngeus—which originate from the styloid process, palate, and auditory tube, respectively. These muscles primarily function to elevate and shorten the pharynx and larynx during swallowing and speech, helping to open the esophageal sphincter and protect the airway. Collectively, this intricate muscular sleeve in the throat serves the vital roles of facilitating swallowing, enabling vocalization, and assisting in respiration by coordinating with the soft palate and epiglottis to direct food away from the nasal cavity and trachea.



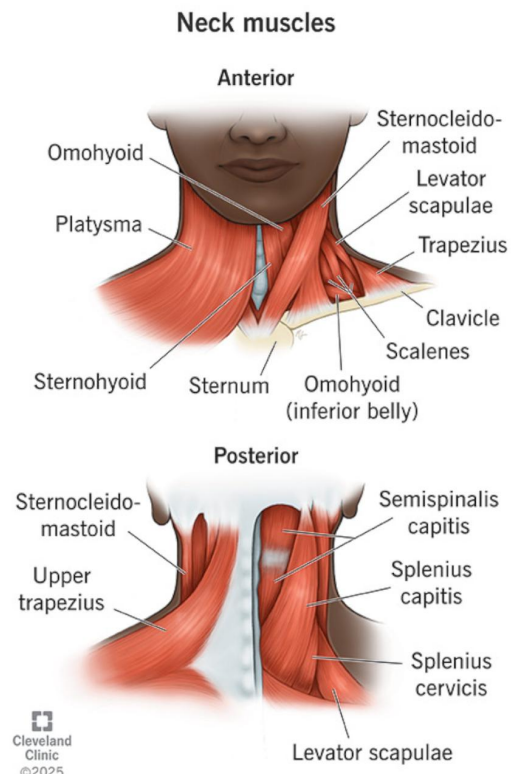
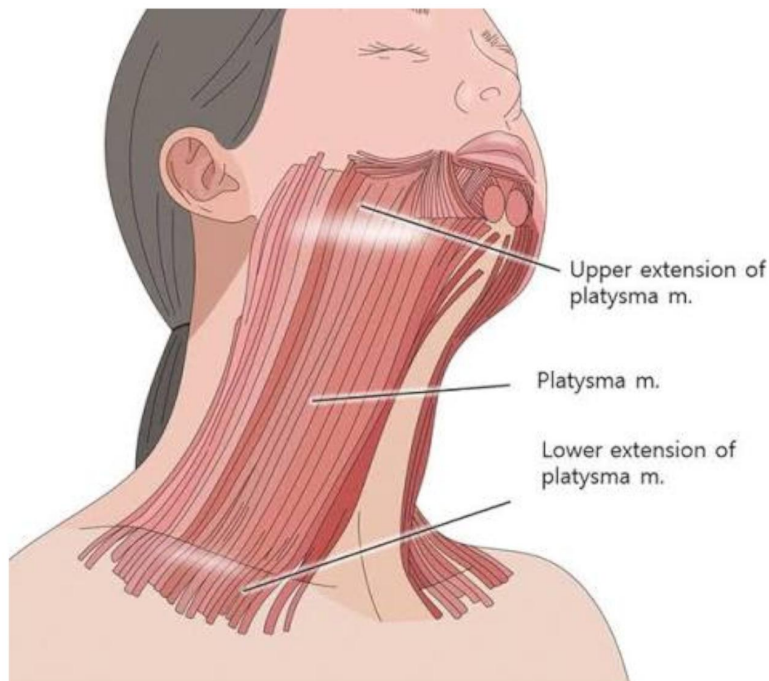
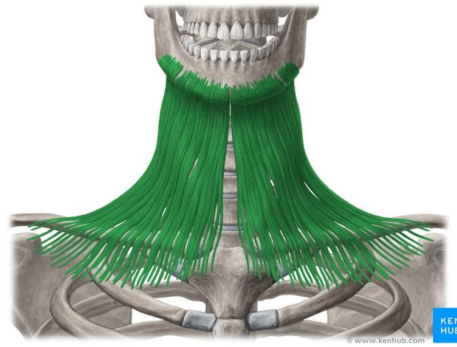
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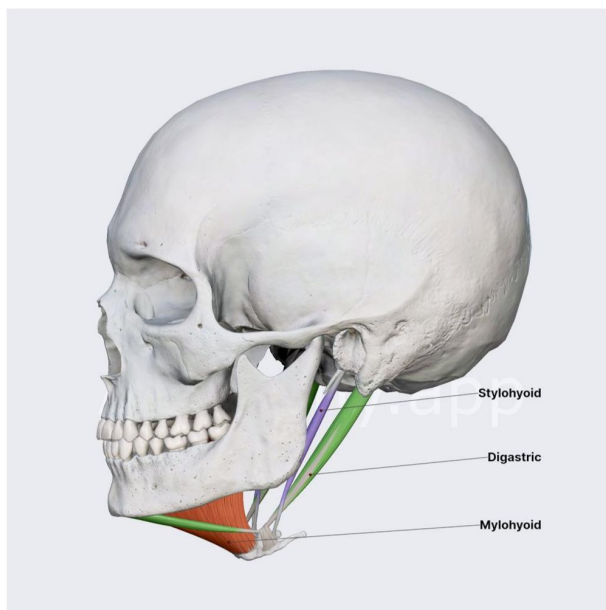
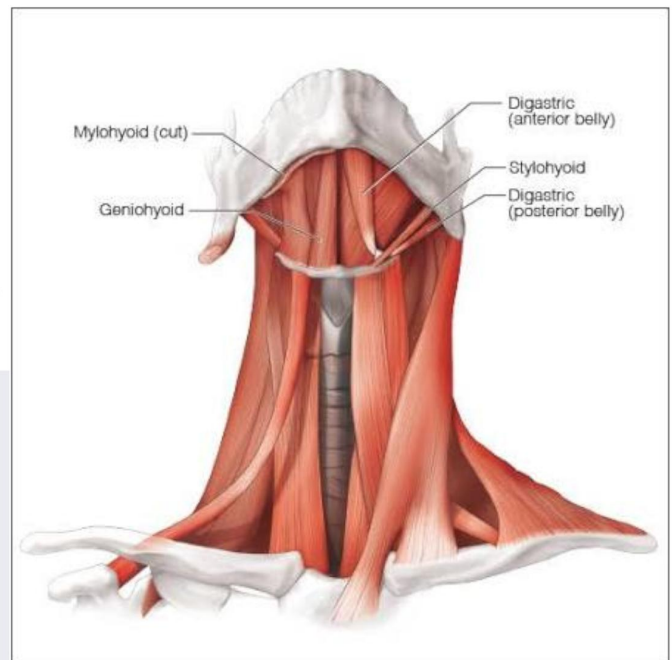
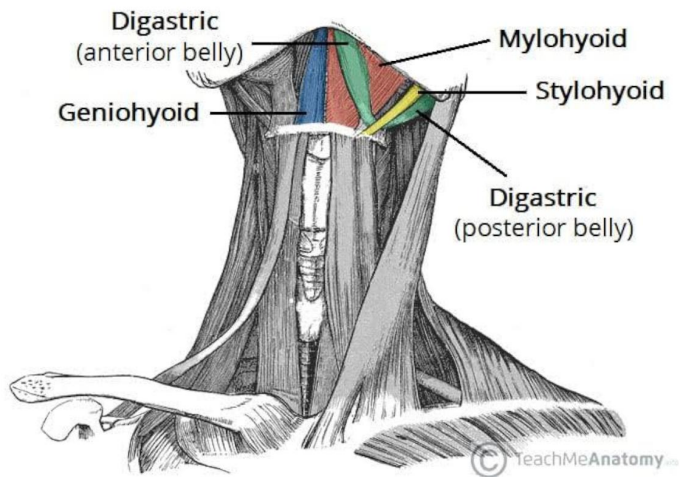
The laryngeal muscles are a complex group of intrinsic and extrinsic muscles located in the anterior neck, forming the structural framework of the voice box, or larynx, which sits at the level of the C3 to C6 vertebrae. The ****extrinsic muscles**** connect the laryngeal cartilages to external structures like the hyoid bone above and the sternum below; they function primarily to elevate or depress the entire larynx during swallowing and vocal pitch changes. In contrast, the ****intrinsic muscles**** have both their origins and insertions entirely within the larynx itself, connecting various cartilages—such as the thyroid, cricoid, and arytenoid cartilages—to one another. These intrinsic muscles are the fine-tuners of the voice, precisely adjusting the tension, length, and position of the vocal folds (vocal cords) to modulate pitch and volume, while also adducting and abducting the folds to close the airway for sound production or open it for breathing, respectively.

mastoid process of temporal bone

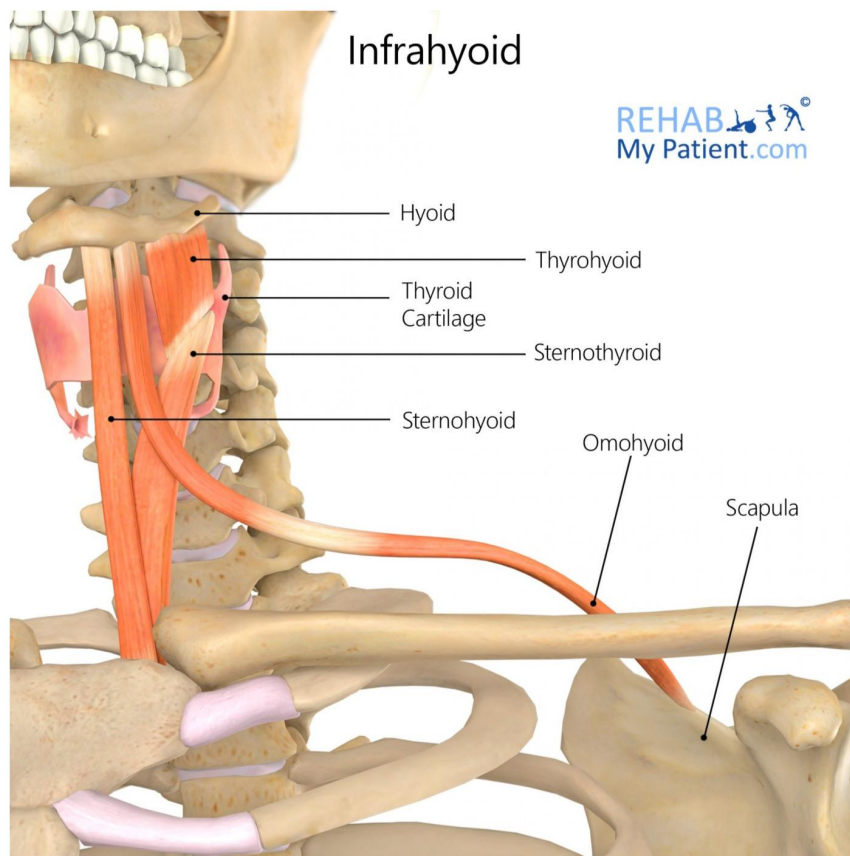


Here is a paragraph describing the platysma and sternocleidomastoideus muscles:

Located in the anterior region of the neck, the platysma and sternocleidomastoideus are superficial muscles with distinctly different functions and connective points. The platysma is a broad, thin sheet of muscle that lies just beneath the skin across the front of the neck. It originates from the fascia covering the upper portions of the pectoralis major and deltoid muscles, and its fibers sweep upward and medially to insert onto the lower border of the mandible and the skin and fascia of the lower face. Its primary action is to depress the mandible and lower lip, while also tensing the skin of the neck, an action that is visible during forceful expiration or expressions of tension, fear, or effort. In stark contrast, the sternocleidomastoideus is a thick, prominent strap-like muscle that forms a distinctive "V" shape on each side of the neck, running diagonally from the mastoid process of the temporal bone (just behind the ear) down to attach to the manubrium of the sternum and the medial third of the clavicle. As the major superficial muscle of the anterolateral neck, its bilateral contraction flexes the cervical spine, drawing the chin toward the chest, while unilateral contraction laterally flexes the head to the same side and rotates it to the opposite side, making it essential for movements like looking over your shoulder.



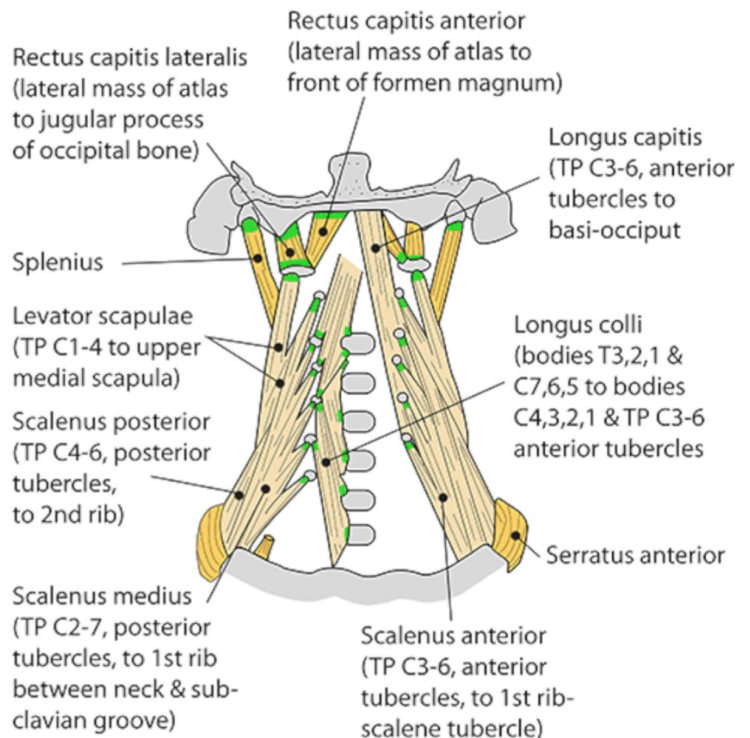
The suprahyoid muscles are a quartet of paired muscles—the digastric, stylohyoid, mylohyoid, and geniohyoid—located in the anterior region of the neck, superior to the hyoid bone. As their name implies, they form a sling-like muscular floor from the mandible down to the hyoid bone, connecting this U-shaped bone to various bony landmarks of the skull and tongue: the digastric attaches from the mastoid process of the temporal bone to the mandible (with its intermediate tendon looping through the hyoid), the stylohyoid runs from the styloid process to the hyoid, the mylohyoid forms a muscular diaphragm from the mylohyoid line of the mandible to the hyoid, and the geniohyoid extends from the inferior mental spine of the mandible to the hyoid. Collectively, their primary actions are to elevate the hyoid bone and the floor of the mouth, which is a critical movement during swallowing (as they help propel the bolus into the pharynx) and during speech for tongue positioning; additionally, when the hyoid is fixed, they assist in depressing the mandible to open the jaw.



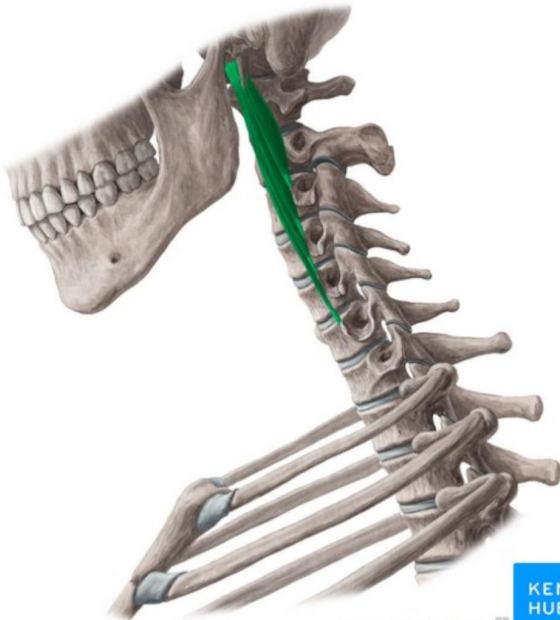
The infrahyoid muscles are a group of four paired strap-like muscles located in the anterior triangle of the neck, inferior to the hyoid bone. This group consists of the sternohyoid, sternothyroid, thyrohyoid, and omohyoid muscles. As their names suggest, they form a muscular sling that connects the hyoid bone superiorly to the sternum and clavicle inferiorly, with the sternothyroid specifically linking the sternum to the thyroid cartilage. Collectively, they function as the primary depressors of the hyoid bone and larynx, acting as antagonists to the suprahyoid muscles. Their key actions are to lower and stabilize the hyoid bone during swallowing and speaking, and to pull the larynx downward following elevation, which is essential for the final phases of deglutition and for phonation.



PREVERTEBRAL MUSCLES



TP= Transverse process



The prevertebral neck muscles are a group of four paired muscles—the **longus colli**, **longus capitis**, **rectus capitis anterior**, and **rectus capitis lateralis**—that form a vertical column along the anterior aspect of the cervical vertebral bodies and transverse processes, deep to the prevertebral fascia. Situated in the deepest layer of the anterior neck, they span from the atlas and axis superiorly down to the third thoracic vertebra inferiorly. Their primary functions are finely tuned actions on the head and neck: the longus colli flexes and slightly rotates the cervical spine, the longus capitis acts as a powerful flexor of the head at the atlanto-occipital joint, and the two rectus capitis muscles provide fine stabilization and assist in nodding and lateral flexion of the head. Collectively, they function as crucial postural stabilizers, maintaining the cervical lordosis, counteracting the pull of stronger extensor muscles, and protecting the spinal cord and vertebral artery by preventing hyperextension and excessive rotation during movements.